# PROPERTIES OF EXPECTATION AND VARIANCE

**DEFINITION FOR X DISCRETE r.v.**

$$E[X] = \sum_{x \in S} x \underbrace{P(X=x)}_{=P_X(x) \text{ pmf of } X}$$

$$Var(X) = \sum_{x \in S} (x - E[X])^2 P(X=x)$$

**X CONTINUOUS r.v.**

$$E[X] = \int_S x \underbrace{P_X(x)}_{\text{pdf of } X} dx$$

$$Var(X) = \int_S (x - E[X])^2 P_X(x) dx$$

$S$: outcome space i.e. set of possible values for $X$

## PROPERTIES OF $E[X]$

① LINEARITY
$$\begin{cases} E[X+Y] = E[X] + E[Y] \\ E[aX] = aE[X] \quad a \in \mathbb{R} \end{cases}$$

② IF $X, Y$ INDEPENDENT $\Rightarrow E[XY] = E[X]E[Y]$

③ MONOTONICITY If $X \geq Y$ then $E[X] \geq E[Y]$

④ $E[g(X)] = \sum_{k=0}^{+\infty} g(k) P(X=k) \quad \left( = \int_0^{+\infty} g(s) P_X(s) ds \right)$

⊛ For $X \geq 0$ $E[X] = \sum_{k=1}^{+\infty} P(X \geq k) \quad \left( = \int_0^{+\infty} P(X \geq s) ds \right)$

EXTRA!

## PROPERTIES OF $Var(X)$

① $Var(X) = E[X^2] - E^2[X]$

recall $Var(-X) = +Var(X)$

$Var(X-Y) = Var(X) + Var(-Y) = Var(X) + Var(Y)$

② IF $X, Y$ INDEPENDENT $Var(X+Y) = Var(X) + Var(Y)$

③ $Var(aX+b) = a^2 Var(X)$ $\quad a, b \in \mathbb{R}$

counter example if $X, Y$ are NOT ind.

$Var(X+Y) \neq Var(X) + Var(Y)$

e.g. $X$ and $Y = -X$

$Var(X-X) = 0$

$\neq$

$Var(X) + Var(-X) = 2Var(X)$

## SKEWNESS

$$SKEW(X) := E\left[\left(\frac{X - E[X]}{SD(X)}\right)^3\right]$$